
A LISP Interpreter for the 8080

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As assembled here, entry is at zero. Most of the code may be anywhere in memory but all LISP data elements (CONS cells and atoms) must be in the first 32K of memory. The interpreter is completely self contained except for terminal I/O. On input from the terminal, all control characters are ignored, and thus may be freely added to improve readability. All input is parsed as literal atoms and S-expressions. An atom is any string of alphanumeric characters not over 16 characters long. Some examples of legal atoms are:

ATOM 934 A47 47A

Some examples of illegal atoms are:

VERYLONGSTRINGOFLETTERS	(More than 16 characters)
AD-P	(Contains non alpha- numeric -)

The only other characters recognized by the system are left and right parentheses, commas, periods and blanks. Consecutive atoms in lists and S-expressions must be separated by at least one of these symbols to be recognized. Blanks may be optionally inserted before left parens and after right parens.

Both input and output routines can handle mixed S-expressions (with dot notation) and list notation. The output routines will always use list notation whenever possible, but will use dotted notation when necessary. E.g., an S-expression input as: ((INTEL.NIL).8080) would print as ((INTEL).8080).

Logical values returned by EQ, ATOM, NULL and EQUAL are T and F. These must be quoted when used as constants in a program. These are also the logical values expected by COND (actually COND only tests for T).

The interpreter is an EVALQUOTE mode interpreter so two S-expressions must be entered for evaluation to occur. Both of these S-expressions must be followed by a blank to allow for the lookahead used in READ. After recognizing the second S-expression, which is a list of arguments to which the first S-expression (typically an atom) will be applied, both of these expressions are effectively quoted, so all atoms in the expressions stand for themselves, not any DEFINED binding. Once the interpreter recognizes the second expression,

it does a new line and types '>>' followed by the result of the computation.

For example, if the following is entered as input:

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DEFINE(((T T)(F F)(FF(LAMBDA(X) (COND((ATOM X)X)
(T(FF(CAR X))))))))
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the interpreter will respond with:

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>> (T F FF)
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If you then enter:

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FF (((A.B).C))
```

the interpreter will respond with:

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>> A
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The following functions are defined as built in subroutines:

ATOM[x] T if the value of X is a literal atom, F otherwise.

NIL is an atom.

NULL[x] T if X has value NIL, F otherwise.

CONS[x,y] Returns a new S-expression whose CAR is X and CDR is Y.

CAR[x] Returns the CAR portion of a CONS cell. The result is undefined when applied to atoms.

CDR[x] Returns the CDR portion of a CONS cell. Undefined for atoms.

EQ[x,y] Returns T if X and Y are the same atom or the same CONS cell, F otherwise.

EQUAL[x,y] Returns T if X and Y are EQ or are congruent trees. Objects which are EQUAL will print identically although they may use different CONS cells, and thus not be EQ.

CADR[x] Same as CAR[CDR[x]]

CADDR[x] Same as CAR[CDR[CDR[x]]]

DEFINE[x] X must be a list of definitions. Each definition is a list whose first element is the name receiving the definition and whose second element is the definition. The value is a list of the names defined.

EVAL[e;a] E is an expression to be evaluated. A is the association list defining the environment for evaluation. The value is the result of this evaluation.

APPLY [fn;x;a] FN is a function name or functional form, X is a list of arguments for the function and A is the association list to be used. The result is the result of applying FN to the arguments in the specified environment.

ASSOC[x;a] X is an atom whose association is fetched from the association list A. The value is the first CONS cell from A of the form CONS [x;value].

0025 23	0035	INX H	HL:=CONS(HL,DE)	0002 C9	0110	REI	0110	CALL ATOM	EQUAL(HL;DE)
0026 72	0036	MOV M,D		0003 2B 00	0111	CALL	0111	CALL ATOM	2-FLAG SET IF EQUAL
0027 28	0037	DCX H		0006 EB	0112	XCHG	0112	XCHG	
0028 28	0038	DCX H		0007 C2 C3 00	0113	JNZ	0113	XNA	
0029 28	0039	DCX H		000A CD 2B 00	0114	CALL	0114	CALL ATOM	
002A C9	0040	RET		000D EB	0115	XCHG	0115	XCHG	
002B 7C	0041	ATOM	ATOM(HL)	000E C0	0116	RNZ	0116	RNZ	NOT BOTH ATOMS, NON-0
002C B6 60	0042	ANI 60H	TEST HIGH ORDER BIT	000F CD 31 00	0117	CALL	0117	CALL EQ	BOTH ATOMS, USE EQ
002E FE 80	0043	CPI 80H	ATOM IF BIT SET	00C2 C9	0118	RET	0118	RET	
0030 C9	0044	RET	2-FLAG IS TRUE	00C3 CD 2B 00	0119	CALL	0119	CALL ATOM	
0031 7C	0045	MOV A,H	EQ(HL;DE)	00C6 EB	0120	XCHG	0120	XCHG	
0032 BA	0046	CMP D	EQUAL IF SAME POINTER	00C7 CA E8 00	0121	JZ	0121	SETNZ	NOT BOTH ATOMS
0033 C0	0047	RNZ		00DE E5	0122	PUSH H	0122	PUSH H	SAVE ROOTS
0034 7D	0048	MOV A,L		00CB D5	0123	PUSH D	0123	PUSH D	
0035 BB	0049	CMP E		00CC CD 09 00	0124	CALL	0124	CALL CAR	
0036 C9	0050	RET		00CF E8	0125	XCHG	0125	XCHG	
0037 E5	0051	ISLIST	2-FLAG SET IF TRUE	00D0 CD 09 00	0126	CALL	0126	CALL CAR	
0038 CD 2B 00	0052	TEST	SAVE, ISLIST(HL)	00D3 CD B3 00	0127	CALL	0127	CALL EQUAL	RECURSIVE ON SUBTREES
003B CA 44 00	0053	CALL ATOM	LOOK FOR ATOMIC	00D6 D1	0128	POP D	0128	POP D	
003E 10 00	0054	CALL CDR	END OF LIST	00D7 E1	0129	POP H	0129	POP H	
0040 C3 38 00	0055	JMP TEST	TRUE IF C(D**N)R=NULL	00D8 C0	0130	RNZ	0130	RNZ	
0044 CD 49 00	0056	CALL NULL	NIL IS ATOM	00D9 E5	0131	PUSH H	0131	PUSH H	BOTH SUBTREES MUST MATCH
0047 E1	0057	POP H		00DA D5	0132	PUSH D	0132	PUSH D	
0048 C9	0058	RET	2-FLAG SET IF TRUE	00DB CD 10 00	0133	CALL	0133	CALL CDR	
0049 3E 80	0059	MVI A,80H	NULL(HL) IF PTR IS 8000H	00DE EB	0134	XCHG	0134	XCHG	
004B BC	0060	CMP H		00DF CD 10 00	0135	CALL	0135	CALL CDR	
004C C0	0061	RNZ	N2=FALSE	00E2 CD B3 00	0136	CALL	0136	CALL EQUAL	RECURSIVE
004D AF	0062	XRA A	A=0	00E5 D1	0137	POP D	0137	POP D	
004E BD	0063	CMP L		00E6 E1	0138	POP H	0138	POP H	
004F C9	0064	RET		00E7 C9	0139	RET	0139	RET	WITH RESULTS OF LAST CALL
0050 E5	0065	OUTPUT		00E8 F6 01	0140	SETNZ	0140	SET NZ FLAG	
0051 CD 49 00	0066	CALL NULL		00EA C9	0141	RET	0141	RET	
0054 C2 5D 00	0067	JNZ NNL		00EB CD 49 00	0142	PAIRLIS	0142	PAIRLIS CALL NULL	PAIRLIS(HL;DE;BC)
0057 21 69 04	0068	LXI H,NILAT	PRINT "NIL"	00EE C2 F4 00	0143	JNZ	0143	FULC	PAIRLIS(X,Y,A)
005A C3 63 00	0069	JMP PAT	GO PRINT ATOM	00F1 60	0144	MOV H,B	0144	MOV H,B	
005D CD 2B 00	0070	CALL ATOM	OUTPUT AN ATOM?	00F2 69	0145	MOV L,C	0145	MOV L,C	
0060 C2 75 00	0071	JNZ NAT		00F3 C9	0146	RET	0146	X=NULL SO RETURN A	
0063 C5	0072	PAT	SAVE	00F4 D5	0147	PUSH D	0147	SAVE Y	
0064 7C	0073	MOV A,H		00F5 E5	0148	PUSH H	0148	PUSH H	
0065 E6 7F	0074	ANI 7FH	CLEAR ATOM FLAG	00F6 D5	0149	PUSH D	0149	PUSH D	
0067 67	0075	MOV H,A		00F7 C2 09 00	0150	CALL	0150	CALL CAR	(X)
0068 46	0076	MOV B,M	ATOM STRING LENGTH	00FA EB	0151	XCHG	0151	XCHG	(Y)
0069 23	0077	ALP		00FB CD 09 00	0152	CALL	0152	CALL CAR	
006A 7E	0078	MOV A,M	GET CHAR	00FE EB	0153	XCHG	0153	XCHG	
006B CD 05 F0	0079	CALL OUTC	PRINT CHAR	00FF CD 16 00	0154	CALL	0154	CALL CONS	(CAR(X),CAR(Y))
006E 05									

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0138 EB      XCHG      JMP      ASSOC2      0185      XCHG      JMP      ASSOC2      0261      RZ      NO ARGS, CALL FN WITH HL=X
0139 C3 17 01      CALL ATOM      0186      CALL ATOM      0262      CPI      2      HOW MANY ARGS
013C CD 2B 00      JNZ      EVAL      0187      JNZ      EVAL      0263      JZ      ARG2      ONLY ONE
0143 C2 53 01      CALL NULL      0188      CALL NULL      0264      JC      ARG1      THREE ARGS, BC=3RD
0142 CD 49 00      RZ      0189      RZ      0265      CALL CADDR      MOV B,H
0145 C8      0190      0266      MOV C,L      HL=X
0146 CD 16 01      0191      0267      MOV L,D
0149 CD 49 00      0192      0268      MOV L,D
014C CA 00 F0      0193      0269      MOV L,E
014F CD 10 00      0194      0270      CALL CADDR
0152 C9      0195      0271      XCHG
0153 E5      0196      0272      CALL CAR
0154 CD 09 00      0197      0273      RET
0157 CD 2B 00      0198      0274      LXI H,TAT+ATOMIC ASSUME TRUE
015A C2 80 01      0199      0275      JZ      APRET      RIGHT
015D D5      0200      0276      LXI H,FAT+ATOMIC NO, FALSE
015E 11 61 84      0201      0277      APRET      RESTORE REGS
0161 CD 31 00      0202      0278      POP B
0164 C2 60 01      0203      0279      RET
0167 D1      0204      0280      NOTDEF      GET FN
0168 E1      0205      0281      MOV D,B      DE=A
0169 CD 06 00      0206      0282      MOV E,C
016C C9      0207      0283      CALL EVAL      (FN,A)
016D 11 3E 84      0208      0284      POP D      RESTORE X
0170 CD 31 00      0209      0285      JMP APPLY      (EVAL(...),X,A)
0173 C2 7F 01      0210      0286      * JMP = CALL, RET
0176 D1      0211      0287      APNA      SAVE FN
0177 E1      0212      0288      CALL CAR      (FN)
0178 CD 10 00      0213      0289      PUSH D      SAVE X
017B C9      0214      0290      LXI D,LAMBDAAT+ATOMIC
017E C9      0215      0291      CALL EQ      (CAR(FN), 'LAMBDA')
017F D1      0216      0292      JNZ      NOTLAM      X
0180 E3      0217      0293      POP D      FN
0181 CD 10 00      0218      0294      POP H
0184 CD 75 02      0219      0295      PUSH H      (FN)
0187 E8      0220      0296      CALL CADDR      (CADDR(FN),X,A)
0188 78      0221      0297      CALL PAIRLIS      (CADDR(FN),X,A)
0189 44      0222      0298      XCHG
018A 67      0223      0299      XTHL
018B 79      0224      0300      CALL CADDR      (FN)
018C 4D      0225      0301      CALL EVAL      (FN)
018D 6F      0226      0302      POP D      GET FN, SAVE CADDR(FN)
018E E3      0227      0303      RET
018F CD 96 01      0228      0304      NOTLAM      LXI D,LABELAT+ATOMIC
0192 50      0229      0305      CALL EQ      (CAR(FN), 'LABEL')
0193 59      0230      0306      JNZ      NOTLB      X
0194 C1      0231      0307      POP D      FN
0195 C9      0232      0308      POP H
0196 CD 2B 00      0233      0309      PUSH H      (FN)
0199 C2 F5 01      0234      0310      CALL CADDR      (FN)
019C D5      0235      0311      CALL CADDR      (FN)
019D E5      0236      0312      CALL CADDR      (FN)
019E 11 37 06      0237      0313      XCHG      X ONTO STACK
01A1 CD 16 01      0238      0314      XTHL
01A4 CD 49 00      0239      0315      XCHG
01A7 CA EF 01      0240      0316      CALL      (CADDR(FN),CADDR(FN))
01AA F1      0241      0317      PUSH D      SAVE CADDR(FN)
01AB CD 10 00      0242      0318      MOV D,B      DE=A
01AE 5D      0243      0319      MOV E,C
01AF 50      0244      0320      CALL CONS      (CONS(...),A)
01B0 CD 10 00      0245      0321      MOV B,H      BC:=CONS(...)
01B3 EB      0246      0322      MOV C,L
01B4 CD 09 00      0247      0323      POP H      CADDR(FN)
01B7 7B      0248      0324      XCHG
01B8 D1      0249      0325      XTHL      SAVE A, GET X
01B9 C5      0250      0326      XCHG
01BB 01 EC 01      0251      0327      CALL APPLY      (CADDR(FN),X,CONS(...))
01BE A7      0252      0328      POP B      A RESTORED
01BF F2 C5 01      0253      0329      RET
01C2 01 E3 01      0254      0330      NOTLB      MVI A,0FFH
01C5 C5      0255      0331      CALL ABEND      ERROR IN LISP PROGRAM
01C6 E5      0256      0332      EVCON      (C,A) IS (HL,DE)
01C7 62      0257      0333      JZ      NOTLAB      ERROR IF C=NULL
01C8 68      0258      0334      PUSH H      SAVE C
01C9 E6 07      0259      0335      CALL CAR      (C)
0199 E6 07      0260      0336      CALL CAR      (CAR(C))

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0256 CD 3C 01      0337 CALL EVAL      (CAAR(C),A)      02E6 02      0412 STAX B
0259 D5           0338 PUSH D          SAVE A          02E7 F1      0413 POP PSW
0260 D1           0339 LXI D,TAT+ATOMIC  02E8 DA ED 02 0414 JC ISLEFT
0261 01           0340 CALL EQUAL      (EVAL(...), 'T') 02E9 23      0415 INX H
0262 01           0341 POP D          C          02EC 23      0416 INX H
0263 01           0342 JNZ CALEC      (C)          02ED 44      0417 ISLEFT MOV B,H
0264 01           0343 CALL CAR      (CAR(C))        02EE 4D      0418 MOV C,L
0265 CD 09 00     0344 CALL CADR      (CADAR(C),A) 02EF CD 09 00 0419 CALL CAR
0266 CD 09 00     0345 CALL EVAL      (C)          02F0 CD 49 00 0420 CALL NULL
0267 CD 3C 01     0346 RET          (C)          02F1 00      0421 JNZ AN
0268 CD 10 00     0347 CALL CDR      (CDR(C),A) - (CALL;REI) 02F2 CD 04 02 0422 PUSH D
0269 CD 10 00     0348 JMP EVCON      (M,A) IS (HL,DE) 02F3 1A      0423 LDAX D
0270 CD 49 02     0349 CALL NULL      EVLIS (NULL;A)=NULL 02F4 C6 04 0424 ADI 4
0271 CD 49 00     0350 RZ          SAVE M          02F5 C2 C4 02 0425 ANI 0FCH
0272 CD 09 00     0351 CALL CAR      (M)          02F6 5F      0426 MOV E,A
0273 CD 09 00     0352 CALL EVAL      (M)          02F7 16 00 0427 MVI D,0
0274 CD 09 00     0353 CALL CDR      (CAR(M))        02F8 D5      0428 CALL GTREM
0275 CD 09 00     0354 CALL EVAL      (M)          02F9 1A      0429 LXI D,0000H
0276 CD 3C 01     0355 CALL CDR      (M)          02FA C6 04 0430 PUSH H
0277 CD 10 00     0356 CALL EVLIS      (CDR(M)) RECURSIVE 02FB 19      0431 DAD D
0278 CD 75 02     0357 XCHG          SAVE A, GET EVAL      02FC E6 FC 0432 SET ATOMIC
0279 E5           0358 CALL CONS      (EVAL(CAR(M)), EVLIS(CDR(M))) 02FD 7D      0433 MOV A,L
0280 E3           0359 POP D          (FN=HL,X=DE,NULL) 02FE 02      0434 STAX B
0281 CD 16 00     0360 RET          (C)          02FF 03      0435 INX B
0282 D1           0361 EVALQUOTE PUSH H      0300 02      0436 MOV A,H
0283 E5           0362 LHLD EVOAL      (M)          0301 00 0437 STAX B
0284 2A B4 04     0363 MOV B,H          (M)          0302 02      0438 POP H
0285 44           0364 MOV C,L          (M)          0303 00 0439 POP D
0286 40           0365 POP H          (M)          0304 11 00 0440 PUSH H
0287 E1           0366 CALL APPLY      (FN=HL,X=DE,NULL) 0305 47      0441 LDAX D
0288 CD 96 01     0367 CALL CRLF      (C)          0306 7C      0442 MOV B,A
0289 CD 21 F0     0368 LXI H,NILAT      (C)          0307 03      0443 MOV M,A
0290 3E 3E        0369 CALL OUTC      (C)          0308 7C      0444 COPYAT INX H
0291 CD 09 F0     0370 CALL OUTC      (C)          0309 13      0445 INX D
0292 CD 09 F0     0371 CALL OUTC      (C)          0310 1A      0446 LDAX D
0293 4D           0372 CALL OUTPUT      (C)          0311 1A      0447 MOV M,A
0294 E1           0373 RET          (C)          0312 D1      0448 DCR B
0295 CD 50 00     0374 MAKEATOM PUSH D      0313 05      0449 JNZ COPYAT
0296 D5           0375 PUSH PSW      (C)          0314 1A      0450 POP H
0297 D5           0376 XCHG          (C)          0315 47      0451 MOV A,H
0298 E8           0377 LXI H,NILAT      (C)          0316 77      0452 ORI 80H
0299 21 69 04     0378 CALL ACOMP      (C)          0317 67 80 0453 MOV H,A
0300 03           0379 JNZ TREES      (C)          0318 13      0454 ACOMP
0301 02 BB 02      0380 LXI H,8000H      (C)          0319 1A      0455 PUSH H
0302 21 00 80      0381 MAKEOUT POP PSW      (C)          0320 7C      0456 PUSH D
0303 01           0382 POP D          (C)          0321 F6 80 0457 MOV A,H
0304 09           0383 RET          (C)          0322 67 7F 0458 ANI 7FH
0305 01           0384 LHL NAMETREE      (C)          0323 1A      0459 MOV H,A
0306 09           0385 CALL AN          (C)          0324 1A      0460 LDAX D
0307 2A 2F 04     0386 JMP MAKEOUT      (C)          0325 05      0461 MOV C,A
0308 04           0387 CALL AN          (C)          0326 05      0462 MOV B,M
0309 07           0388 MOV B,H          (C)          0327 23      0463 COMPLP INX H
0310 02           0389 MOV C,L          (C)          0328 13      0464 INX D
0311 04           0390 CALL CDR      (C)          0329 1A      0465 LDAX D
0312 05           0391 PUSH PSW      (C)          0330 96      0466 SUB M
0313 00           0392 CALL CAR      (C)          0331 4F      0467 JNZ NEOL
0314 00           0393 RZ          (C)          0332 05      0468 DCR B
0315 00           0394 MOV L,C          (C)          0333 44      0469 JZ DONE1
0316 00           0395 CALL CDR      (C)          0334 0D      0470 DCR C
0317 00           0396 PUSH PSW      (C)          0335 2F 03 0471 JNZ COMPLP
0318 00           0397 CALL NULL      (C)          0336 80      0472 STA B
0319 00           0398 JNZ ISLR      (C)          0337 37      0473 SET NZ
0320 00           0399 JNZ ISLR      (C)          0338 01      0474 HL LONGER, SET '>'
0321 00           0400 SAVE          (C)          0339 01      0475 POP B
0322 00           0401 DE=NIL          (C)          0340 D1      0476 POP D
0323 00           0402 (NIL;NIL) ADDING TO TREE 0341 01      0477 POP H
0324 00           0403 INSERT INTO EXISTING TREE 0342 01      0478 RET
0325 00           0404 CHANGE LINK      (C)          0343 09      0479 DCR C
0326 00           0405 POP D          (C)          0344 0D      0480 JMP NEOL
0327 00           0406 INX B          (C)          0345 C3 40 03 0481 NEXTCHAR CALL INCH
0328 00           0407 INX B          (C)          0346 F0 06 F0 0482 CPI
0329 00           0408 MOV A,L          (C)          0347 20 06 F0 0483 JZ NEXTCHAR
0330 00           0409 STAX B          (C)          0348 DA 48 03 0484 STA CURCH
0331 00           0410 INX B          (C)          0349 9C 04 0485 RET
0332 00           0411 MOV A,H          (C)          0350 32 9C 04 0486 CALL TAKELP
0333 00           0412 2=TRUE GET '(' IF AVAIL 0351 C9      0487
0334 00           0413 0354 CD C4 03

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0357 C2 9C 03      JNZ S139      JUMP IF NOT THERE
035A CD D7 03      CALL TAKEKP  GET ' ) ' IF THERE
035D C2 64 03      JNZ S127      JUMP IF NOT THERE
0360 21 00 80      LXI H,8000H  NIL IS ( )
0363 C9            RET
0364 D5            PUSH D
0365 CD 54 03      CALL INPUT   RECURSIVE CALL
0368 EB            XCHG      SAVE 'INPUT'
0369 CD CF 03      CALL TAKEDOT TRY FOR ' ' (SEXP)
036C D2 75 03      JNZ S130      JUMP IF LIST NOVATION
036F CD 54 03      CALL INPUT   GET CDR SIDE
0372 C3 90 03      JMP S135      GO GET ' ) '
0375 CD ED 03      CALL TAKEBL  LIST:GET SEP ' ' OR ' '
0378 C2 81 03      JNZ S132      NO SEPARATOR
037B CD A9 03      CALL INPTL   GET REST OF LIST
037E C3 90 03      JMP S135      THEN ' ) '
0381 CD DF 03      CALL CTRP    SEE IF ' ) ', BUT DONT TAKE
0384 C2 80 03      JNZ S134      JUMP S134
0387 21 00 80      LXI H,8000H  NIL
038A C3 90 03      JMP S135      JUMP S135
038D CD A9 03      CALL INPTL   CALL TAKEKP  TAKE ' ) '
0390 CD D7 03      EQU NOTLB    ILLEGAL STRING
0393 C2 44 02      JNZ BADIN    JNZ BADIN
0396 EB            XCHG
0397 CD 16 00      CALL CONS    POP D
039A D1            POP D
039B C9            RET
039C CD E5 03      CALL CTATOM  RET
039F C2 44 02      JNZ S139      JNZ BADIN
03A2 CD 11 04      CALL TAKEATOM PUT ATOM TEXT IN ATXT
03A5 CD A7 02      CALL MAKEATOM GET POINTER FOR IT
03A8 C9            RET
03AA CD DF 03      CALL CTRP    JNZ S144
03AC C2 B3 03      JNZ S144      LXI H,8000H  NIL
03AF 21 00 80      RET
03B2 C9            RET
03B3 D5            PUSH D
03B4 CD 54 03      CALL INPUT   CALL INPUT
03B7 EB            XCHG
03B8 CD ED 03      CALL TAKEBL  CALL TAKEBL
03BB CD A9 03      CALL INPTL   CALL INPTL
03BE EB            XCHG
03BF CD 16 00      CALL CONS    CALL CONS
03C2 D1            POP D
03C3 C9            RET
03C4 3A 9C 04      TAKELP LDA   CTRP
03C7 FE 28         CPI      CURI
03C9 C0            RET
03CA CD 48 02      CALL NEXTCHAR
03CD AF            XRA A
03CE C9            RET
03CF 3A 9C 04      TAKEDOT LDA  CURI
03D2 FE 2E         CPI      CURI
03D4 C3 C9 03      JNZ TAKANY   JNZ TAKANY
03D7 3A 9C 04      TAKERP LDA   CURI
03DA FE 29         CPI      CURI
03DC C3 C9 03      JMP          CTRP
03DF 3A 9C 04      LDA CURI
03E2 FE 29         CPI      CURI
03E4 C9            RET
03E5 3A 9C 04      CTATOM LDA   CURI
03E8 FE 30         CPI      CURI
03EA D8            RC
03EB AF            XRA A
03EC C9            RET
03ED 3A 9C 04      TAKEBL LDA   CURI
03F0 FE 20         CPI      CURI
03F2 CA FC 03      JZ TAKING    JZ TAKING
03F5 DA FC 03      JC TAKING    JC TAKING
03F8 AF            XRA A
03F9 FE 2C         CPI      CURI
03FB C0            RNZ
03FC CD 48 03      CALL NEXTCHAR
03FF 3A 9C 04      LDA CURI
0402 FE 20         CPI      CURI
0404 C2 9C 03      JNZ S139      JUMP IF NOT THERE
0407 DA FC 03      CALL TAKEKP  GET ' ) ' IF THERE
040A FE 2C         CPI      CURI
040C CA FC 03      JZ TAKING    JZ TAKING
040F AF            XRA A
0410 C9            RET
0411 C5            TAKEATOM PUSH B SAVE
0412 06 00 00      MVI B,0      INITIAL ATOM LENGTH
0414 21 9E 04      LXI H,ATXT   LXI H,ATXT
0417 CD E5 03      TATL         CALL CTATOM
041A C2 29 04      JNZ FINISHT  JNZ FINISHT
041D 3A 9C 04      LDA CURI     LDA CURI
0420 77           MOV M,A   MOV M,A
0421 23           INR H      INR H
0422 04           CALL NEXTCHAR
0423 CD 48 03      JNZ S139      JNZ BADIN
0426 C3 17 04      FINISHT LXI  H,ATL
042C 70           MOV M,B   MOV M,B
042D C1           POP B      POP B
042E C9           RET
042F 00 00 00     EQU 0      ATOM PTR FLAG
0431 04           EQU 4      ATOM PTR FLAG
0432 41 54         DW 'TA'   DW 'TA'
0434 4F 4D         DW 'MO'   DW 'MO'
0436 03           DW 3       DW 3
0437 43 41         DW 'AC'   DW 'AC'
0439 52           DW 'R'     DW 'R'
043A 03           DW 3       DW 3
043B 43 44         DW 'DC'   DW 'DC'
043D 52           DW 'R'     DW 'R'
043E 04           DW 4       DW 4
043F 43 4F         DW 'OC'   DW 'OC'
0441 4E 44         DW 'DN'   DW 'DN'
0443 04           DW 4       DW 4
0444 43 4F         DW 'OC'   DW 'OC'
0446 4E 53         DW 'SN'   DW 'SN'
0448 06           DW 6       DW 6
0449 44 45         DW 'ED'   DW 'ED'
044B 46 49         DW 'IF'   DW 'IF'
044D 4E 45         DW 'EN'   DW 'EN'
044F 02           DW 2       DW 2
0450 45 51         DW 'EQ'   DW 'EQ'
0452 01           DW 1       DW 1
0453 46           DW 'F'     DW 'F'
0454 05           DW 5       DW 5
0455 4C 41         DW 'AL'   DW 'AL'
0457 42 45         DW 'EB'   DW 'EB'
0459 4C           DW 'L'     DW 'L'
045A 06           DW 6       DW 6
045B 4C 41         DW 'AL'   DW 'AL'
045D 4D 42         DW 'BM'   DW 'BM'
045F 44 41         DW 'AD'   DW 'AD'
0461 05           DW 5       DW 5
0462 51 55         DW 'UQ'   DW 'UQ'
0464 4F 54         DW 'TO'   DW 'TO'
0466 45           DW 'E'     DW 'E'
0467 01           DW 1       DW 1
0468 54           DW 'H'     DW 'H'
0469 03           DW 3       DW 3
046A 4E 49         DW 'IN'   DW 'IN'
046C 4C           DW 'L'     DW 'L'
046D 05           DW 5       DW 5
046E 43 41         DW 'AC'   DW 'AC'
046F 44 44         DW 'DD'   DW 'DD'
0472 52           DW 'R'     DW 'R'
0473 04           DW 4       DW 4
0474 43 41         DW 'AC'   DW 'AC'
0476 44 52         DW 'RD'   DW 'RD'
0478 04           DW 4       DW 4
0479 4E 55         DW 'UN'   DW 'UN'
047B 4C 4C         DW 'LL'   DW 'LL'
047D 05           DW 5       DW 5
047E 45 51         DW 'QE'   DW 'QE'
0480 55 41         DW 'AU'   DW 'AU'

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05CA CA D0 05	0789	JZ	HIEQ	JUMP IF LENGTHS CLOSE	064D 4F 06	0862	DW	AF18
05CD DA BA 05	0790	JC	LP	NO, TRY NEXT	064F 87 06	0863	DW	AF19
05D0 47	0791	HIEQ	MOV B,A		0651 53 06	0864	DW	AF21
05D1 79	0792	MOV	MOV A,C		0653 8B 06	0865	DW	AF22
05D2 C2 E9 05	0793	JN2	BIGH	FQE MUCH BIGGER	0655 57 06	0866	DW	AF24
05D5 93	0794	SUB	E		0657 8F 06	0867	DW	AF25
05D6 CA F1 05	0795	JZ	EXACT	FQE SIZE=REQ SIZE	0659 5B 06	0868	DW	AF27
05D9 DA BA 05	0796	JC	LP	WON'T FIT	065B 93 06	0869	DW	AF28
05DC 4F	0797	RETPART	MOV C,A BC	NOW EXCESS	065D 5F 06	0870	DW	AF30
05DD C5	0798	PUSH B		CHANGE FQE LENGTH	065F 97 06	0871	DW	AF31
05DE 09	0799	DAD B		CALCULATE START OF RETURN AR	0661 63 06	0872	DW	AF33
05DF EB	0800	EXIT	XCHG	RESTORE ORIGINAL SP	0663 9B 06	0873	DW	AF34
05E0 2A B0 04	0801	LHLD	SPSAV	SAVE HL	0665 67 06	0874	DW	AF36
05E3 F9	0802	SPHL		'DE' TO RESTORE	0667 9F 06	0875	DW	AF37
05E4 E1	0803	POP H			0669 6B 06	0876	DW	AF39
05E5 C1	0804	POP B			066B A3 06	0877	DW	AF40
05E6 F1	0805	POP PSW			066D 00 80	0878	DW	NIL
05E7 EB	0806	XCHG		SWAP TO RIGHT REGS	066F 36 84	0879	DW	CARAT+ATOMIC
05E8 C9	0807	RET			0671 A7 06	0880	DW	AF2
05E9 93	0808	BIGH	E	BC=BC-DE	0673 3A 84	0881	DW	CDRAT+ATOMIC
05EA D2 DC 05	0809	JNC	RETPART	BC SET IF NO BORROW	0675 AB 06	0882	DW	AF5
05ED 05	0810	DCR B		BORROW	0677 43 84	0883	DW	CONSAT+ATOMIC
05EE C3 DC 05	0811	JMP	RETPART		0679 AF 06	0884	DW	AF8
05F1 F9	0812	EXACT	SPHL		067B 31 84	0885	DW	ATOMAT+ATOMIC
05F2 C1	0813	POP B		GET PTR TO NEXT FQE	067D B3 06	0886	DW	AF11
05F3 2A B2 04	0814	LHLD	PREVFOE	CHAIN BACK	067F 4F 84	0887	DW	EQAT+ATOMIC
05F6 F9	0815	SPHL		TO PREVIOUS	0681 B7 06	0888	DW	AF14
05F7 E1	0816	POP H		INCREMENT SP 2	0683 48 84	0889	DW	DEFINAT+ATOMIC
05F8 C5	0817	PUSH B		CLOSE CHAIN	0685 BB 06	0890	DW	AF17
05F9 C3 DF 05	0818	JMP	EXIT		0687 6D 84	0891	DW	CADDAT+ATOMIC
05FC C3 00 F0	0819	NOSPAC	JMP	ABEND	0689 BF 06	0892	DW	AF20
05FF	0820	NIL	EQ	+ATOMIC	068B 73 84	0893	DW	CDRAT+ATOMIC
05FF CD 49 00	0821	DEFNE	CALL	NIL	068D C3 06	0894	DW	AF23
0603 E5	0822	RZ			068F 78 84	0895	DW	NULLAT+ATOMIC
0604 CD 10 00	0823	PUSH H		SAVE	0691 C7 06	0896	DW	AF26
0607 CD FF 05	0824	CALL	CDR		0693 7D 84	0897	DW	EQULAT+ATOMIC
060A E3	0825	CALL	DEFNE	RECURSE	0695 CB 06	0898	DW	AF29
060B CD 16 06	0826	XTHL			0697 83 84	0899	DW	PAIRAT+ATOMIC
060E EB	0827	CALL	DEF		0699 CF 06	0900	DW	AF32
060F E3	0828	XCHG			069B 8B 84	0901	DW	ASOCAT+ATOMIC
0610 EB	0829	XTHL			069D D3 06	0902	DW	AF35
0611 CD 16 00	0830	XCHG			069F 91 84	0903	DW	AF38
0614 D1	0831	CALL	CONS		06A1 D7 06	0904	DW	AF38
0615 C9	0832	POP D			06A3 96 84	0905	DW	AF40
0616 CD 09 00	0833	RET			06A5 DB 06	0906	DW	AF41
0619 E5	0834	DEF			06A7 09 00	0907	DW	CAR
061A D5	0835	PUSH H			06A9 01 00	0908	DW	1
061B 54	0836	PUSH D			06AB 10 00	0909	DW	CDR
061C 5D	0837	MOV D,H		SAVE ARG	06AD 01 00	0910	DW	1
061D CD 06 00	0838	MOV E,L			06AF 16 00	0911	DW	CONS
0620 EB	0839	CALL	CADR	FN DEF	06B1 02 00	0912	DW	2
0621 CD 09 00	0840	XCHG			06B3 2B 00	0913	DW	ATOM
0624 CD 16 00	0841	CALL	CAR		06B5 81 00	0914	DW	81H
0627 EB	0842	CALL	CONS	FOR A-LIST	06B7 31 00	0915	DW	EQ
0628 2A B4 04	0843	XCHG			06B9 82 00	0916	DW	82H
062B EB	0844	LHLD	EVQAL	CURRENT LIST	06BB FF 05	0917	DW	DEFNE
062C CD 16 00	0845	XCHG			06BD 01 00	0918	DW	1
062F 22 B4 04	0846	CALL	CONS		06BF 03 00	0919	DW	CADDR
0632 D1	0847	SHLD	EVQAL		06C1 01 00	0920	DW	1
0633 E1	0848	POP D			06C3 06 00	0921	DW	CADR
0634 C3 09 00	0849	POP H			06C5 01 00	0922	DW	1
0637 6F 06	0850	JMP	CAR	=CALL,RET	06C7 49 00	0923	DW	NULL
0639 3B 06	0851	ASMFN	DW	ASSOC LIST OF ASSM FNS	06C9 81 00	0924	DW	81H
063B 73 06	0852	DW	AF3		06CB B3 00	0925	DW	EQUAL
063D 3F 06	0853	DW	AF4		06CD 82 00	0926	DW	82H
063F 77 06	0854	DW	AF6		06CF EB 00	0927	DW	PAIRLIS
0641 43 06	0855	DW	AF7		06D1 03 00	0928	DW	3
0643 7B 06	0856	DW	AF9		06D3 16 01	0929	DW	ASSOC
0645 47 06	0857	DW	AF10		06D5 02 00	0930	DW	2
0647 7F 06	0858	DW	AF12		06D7 3C 01	0931	DW	EVAL
0649 4B 06	0859	DW	AF13		06D9 02 00	0932	DW	2
064B 83 06	0860	DW	AF15		06DB 96 01	0933	DW	APPLY
	0861	AF15	DW	AF16	06DD 03 00	0934	DW	3